

COUNTER FLOW COOLING TOWERS



***“EMERGING GLOBAL LEADER IN MANUFACTURING AND
SUPPLY OF COOLING TOWERS AND ALLIED UTILITY*”**

BLUECHIP make **Counter Flow Cooling Tower** is a cooling system where the flow of air and water moves in opposite directions, enabling efficient heat exchange. Hot water enters at the top and flows downward through the fill media, while air is drawn upward from the bottom, directly opposing the water flow. Counterflow cooling towers are valued for their superior cooling efficiency and space-saving design, making them ideal for high-performance cooling needs in industrial and commercial settings.



KEY FEATURES:

- **AIR FLOW PATTERN:** The counterflow of air and water maximizes contact, enhancing heat transfer efficiency.
- **PRESSURIZED WATER DISTRIBUTION:** Typically uses spray nozzles to distribute water over the fill, requiring higher water pressure.
- **COMPACT DESIGN:** Requires less ground space compared to crossflow cooling towers.

ADVANTAGES:

- **HIGH EFFICIENCY:** Counterflow design allows for better air-to-water contact, leading to improved cooling performance.
- **SMALLER FOOTPRINT:** Ideal for locations with limited space.
- **REDUCED WATER EXPOSURE:** Enclosed water distribution reduces contamination risks.

APPLICATIONS:

- Used in industries like power plants, petrochemical facilities, and HVAC systems requiring efficient cooling in space-constrained environments.

CONSIDERATIONS:

- **MAINTENANCE ACCESS:** The compact design can make it harder to access internal components.
- **ENERGY DEMAND:** Pressurized spray systems may require higher energy input.

WHY CHOOSE A BLUECHIP MAKE CROSSFLOW COOLING TOWER?

- **High Efficiency:** Superior thermal performance due to advanced counterflow design.
- **Compact Design:** Space-efficient towers for installations with limited area.
- **Durability:** Built with premium-grade materials for long-lasting performance.
- **Low Energy Consumption:** Designed to reduce operational costs and carbon footprint.
- **Easy Maintenance:** Accessible features for quick inspections and servicing.

TECHNICAL SPECIFICATIONS:

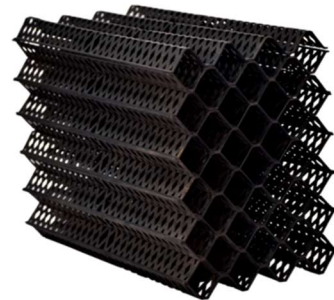
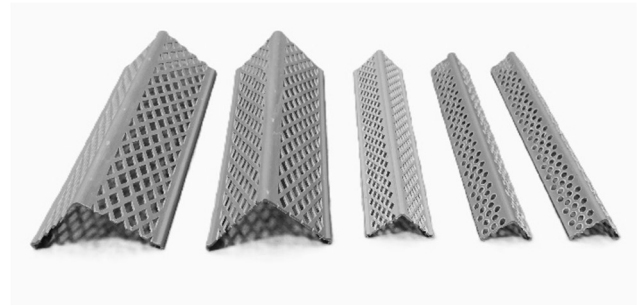
Specification	Details
Cooling Capacity	Customizable (5 – 10000+ TR) per cell capacity
Flow Rate	3 m3/h to 20,000 m3/h per cell capacity
Material Options	Treated Timber, Pultruded FRP, Steel, RCC
Noise Levels	Low-noise operation (<85 dBA)
Water Drift Loss	<0.001%
Operating Temperature Range	Suitable for extreme conditions (3 Degrees to 75 Degrees)

INNOVATING COOLING TECHNOLOGY FOR A SUSTAINABLE FUTURE

COMPONENTS THAT BUILT A BLUECHIP COUNTERFLOW COOLING TOWER

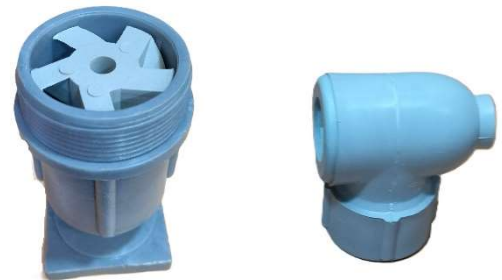
1. FILL MEDIA

- **MATERIAL:** High-performance PVC or PP material for enhanced durability.
- **FUNCTION:** Maximizes heat exchange by creating a large contact area between water and air.
- **DESIGN VERSATILITY:** Compatible with various types of fills, including:
 - **HONEYCOMB FILLS:** Provides high surface area for efficient cooling.
 - **TRICKLE FILLS:** Ideal for low-fouling applications.
 - **V BAR FILLS:** Enhances water distribution for optimal performance.
 - **SPLASH FILLS:** Suitable for applications requiring higher turbulence.
- **CLOG-RESISTANT DESIGN:** Engineered to prevent blockages and maintain consistent operation.



2. WATER DISTRIBUTION SYSTEM

- **PRESSURIZED SPRAY NOZZLES:** Ensures uniform water distribution over the fill media for optimal cooling.
- **PIPING:** Made from corrosion-resistant materials to maintain system integrity.
- **AVAILABLE PIPING OPTIONS:** GI / PVC / PP / PPRC





3. DRIFT ELIMINATORS

- **PURPOSE:** Prevents water loss by capturing droplets carried with airflow.
- **MATERIAL:** High-efficiency, corrosion-resistant materials ensure long-term performance.

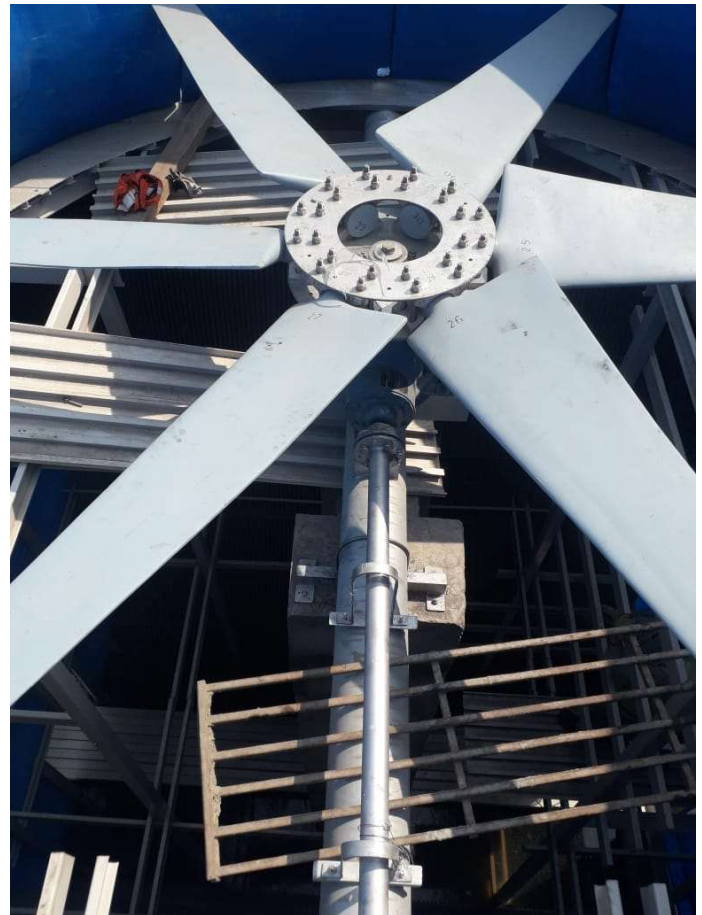
4. AIR INLET LOUVERS

- **MATERIAL:** FRP louvers protect internal components and improve airflow.
- **FUNCTION:** Prevents debris entry and minimizes water splash-out.



5. FANS AND MOTOR ASSEMBLY

- **FANS:** High-efficiency axial fans ensure smooth and quiet operation.



- **MOTORS:** Available in Direct Drive and Geared Driven systems for enhanced reliability and efficiency.

A. DIRECT DRIVE MOTORS

- **ADVANTAGES:**
 - **REDUCED MECHANICAL COMPLEXITY:** Fewer moving parts compared to belt drive systems. Ideal for Small cooling towers
 - **HIGHER ENERGY EFFICIENCY:** Direct connection eliminates power losses from belts and gears.
 - **LOWER MAINTENANCE:** No belts to adjust or replace, elimination of gears and complex oil schedules, reducing downtime.



B. GEARED DRIVEN MOTORS

- **ADVANTAGES:**
 - **HIGH TORQUE CAPABILITY:** Suitable for larger cooling towers with heavy loads.
 - **COMPACT DESIGN:** Efficient power transfer ensuring applicability in smaller footprint.
 - **DURABLE AND ROBUST:** Designed for demanding industrial applications.

COMPARISON WITH BELT DRIVE SYSTEMS

- **EFFICIENCY:** Direct drive and geared systems deliver higher efficiency by reducing energy losses inherent in belt-driven systems.
- **MAINTENANCE:** Eliminates the need for frequent belt adjustments or replacements.
- **LONGEVITY:** Reduces wear and tear on components, extending the lifespan of the motor and fan assembly.

6. STRUCTURAL FRAME

- **MATERIAL:** Treated Timber, Pultruded FRP, Galvanized Steel, Stainless Steel for long life corrosion resistance and structural integrity.
- **DESIGN:** Reinforced structure to withstand mechanical stress and environmental conditions.



SAFETY FEATURES

- **FAN GUARD:** Protects against accidental contact with rotating components.
- **HANDRAIL AND LADDER:** Ensures safe access for maintenance and inspections.
- **EMERGENCY CAGE:** Ladder with safety cage for enhanced worker protection.
- **ANTI-SLIP FANDECK:** Facilitates safe movement during maintenance.



MONITORING INSTRUMENTS

- **OIL LEVEL SWITCH:** Monitors motor oil levels to ensure optimal performance.
- **VIBRATION LIMIT SWITCH:** Detects abnormal vibrations to prevent potential damage.



COOLING TOWER ACCESSORIES

- **ANCHOR BOLTS:** Securely fasten the cooling tower to its foundation, ensuring stability during operation.
- **DOUBLE DECK STAIRCASE:** Provides safe and efficient access to multiple levels of the cooling tower.
- **DERRICK ARM ARRANGEMENT:** Assists in lifting heavy components for maintenance or replacement, improving operational efficiency.



APPLICATIONS

BLUECHIP Counter Flow Cooling Towers are versatile and ideal for:

INDUSTRIAL APPLICATIONS

- Power plants
- Petrochemical and refining industries
- Steel and aluminum manufacturing
- Textile processing

COMMERCIAL APPLICATIONS

- HVAC systems in commercial buildings
- Data centers and IT infrastructure cooling
- Hospitals and healthcare facilities

SPECIALIZED INDUSTRIES

- Food and beverage manufacturing
- Pharmaceutical and chemical production
- Automotive and aerospace sectors

CUSMER SUPPORT AND SERVICES

- **CONSULTATION AND DESIGN:** Custom cooling solutions tailored to your specific needs.
- **INSTALLATION ASSISTANCE:** Expert installation and commissioning support.
- **MAINTENANCE PLANS:** Comprehensive service plans to ensure optimal performance.
- **24 / 7 TECHNICAL SUPPORT:** Our team is always ready to assist you.

CONTACT US TODAY

Maximize your cooling efficiency and reduce operational costs with BLUECHIP's Counter Flow Cooling Towers. Let us help you achieve unparalleled performance for your industrial and commercial cooling needs.

PHONE: +91 9653150937 / 9321780765

EMAIL: office@bluechip-industries.com

WEBSITE: www.bluechipindustries.com

ADDRESS: Plot No - G88, MIDC Tarapur, MIDC Road, Behind Hotel RC, Boisar, Dist - Palghar, Maharashtra - 401506.



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